

Security Investigation Report

TOR Network Sign-In Detection via KQL

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1. Purpose

This report documents the rationale, methodology, and implementation for capturing and retaining user sign-in logging data, specifically as it relates to detecting authentication attempts originating from TOR (The Onion Router) network exit nodes.

The objective is to identify potentially suspicious or anonymized access attempts to organizational resources via Microsoft Entra ID (formerly Azure Active Directory), to then forward them to our AI model for advanced determination of phishing attempts.

2. Background on TOR Network Risks

The TOR network is a privacy-focused anonymization service that routes internet traffic through a series of encrypted relays, ultimately exiting through publicly listed exit nodes. While TOR has legitimate use cases, it is also widely used by malicious actors to:

- Conceal the true origin of unauthorized access attempts.
- Bypass IP-based geo-restrictions and conditional access policies.
- Conduct credential stuffing or brute-force attacks without attribution.
- Exfiltrate data or maintain persistence while evading detection.

Sign-ins from TOR exit nodes are considered a high-severity signal in most security frameworks and are flagged by Microsoft as a risk indicator within Entra ID Identity Protection.

3. Logging Data Captured

3.1 Data Source

Authentication events are captured from the `EntraIdSignInEvents` table, which logs all sign-in activity across the Microsoft Entra ID tenant, including:

- User principal name (UPN)
- Source IP address
- Timestamp of authentication event

- Authentication result (success/failure)
- Device, application, and location metadata

3.2 TOR Exit Node Reference List

Known TOR exit node IP addresses are dynamically retrieved from the official TOR Project bulk exit list, hosted at:

```
https://check.torproject.org/torbulkexitlist
```

This list is maintained by the TOR Project and updated continuously. Using `externaldata()` in KQL ensures the query always references the most current list of known TOR exit nodes at the time of execution, without requiring manual list maintenance.

4. Detection Methodology

The detection pipeline consists of two stages:

Stage 1	Scrape and ingest the current TOR exit node IP list from the TOR Project's public bulk export endpoint using KQL's <code>externaldata()</code> function.
Stage 2	Cross-reference all sign-in events in the <code>EntraIdSignInEvents</code> table against the ingested TOR IP list to surface any matching authentication attempts.

This approach is fully automated and requires no manual IP list curation, ensuring detection remains current as TOR exit nodes change over time.

5. KQL Detection Query

The following KQL (Kusto Query Language) query was written and executed within Microsoft Sentinel / Defender XDR to perform the TOR sign-in detection:

```
let TorExitNodes = externaldata(IPAddress: string)
    [@"https://check.torproject.org/torbulkexitlist"]
    with (format="txt");
EntraIdSignInEvents
| where tostring(IPAddress) in (TorExitNodes)
```

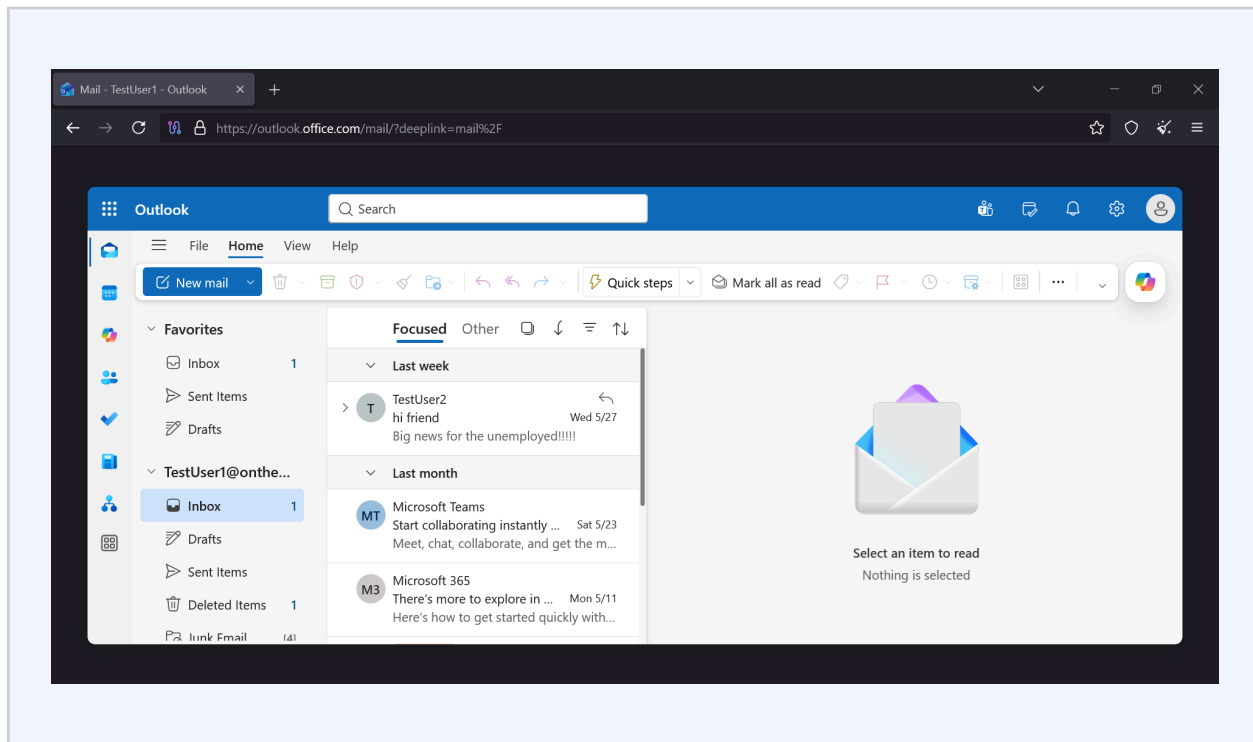
5.1 Query Explanation

externaldata()	Dynamically loads the TOR exit node list at query runtime from the TOR Project URL, parsing it as a plain-text list of IP addresses.
EntraldSignInEvents	The primary table contains all Entra ID sign-in log events across the tenant.
where IPAddress in (...)	Filters sign-in events to only those where the source IP matches a known TOR exit node.

6. Proof of Concept

To validate the detection capability, a controlled test was conducted simulating a sign-in from a TOR exit node IP address. The following evidence is included in this report:

- Screenshot: Successful sign-in from TOR browser/exit node IP.
- Screenshot: KQL query output in Microsoft Sentinel confirming detection of the TOR-sourced sign-in event.



Query

Run query

Create detection rule

30,000 (Default)

Last 7 days

```

1 let TorExitNodes = externaldata(IPAddress: string)
2   [@"https://check.torproject.org/torbulkexitlist"]
3   with (format="txt");
4 EntraIdSignInEvents
5 | where toString(IPAddress) in (TorExitNodes)

```

Results

Query history

Getting started

Export

Show empty columns

14 items

Search

00:02.499

Low

	Timestamp	Application	ApplicationId	LogonType	EndpointCall	ErrorCode
☐	> Jun 1, 2026 10:27:26	One Outlook Web	9199bf20-a13f-4107-85dc	["interactiveUser"]	OAuth2:Token	0
☐	> Jun 1, 2026 10:23:25	One Outlook Web	9199bf20-a13f-4107-85dc	["interactiveUser"]	OAuth2:Token	0
☐	> Jun 1, 2026 10:23:25	One Outlook Web	9199bf20-a13f-4107-85dc	["interactiveUser"]	OAuth2:Token	0
☐	> Jun 1, 2026 10:23:02	One Outlook Web	9199bf20-a13f-4107-85dc	["interactiveUser"]	OAuth2:Token	0
☐	> Jun 1, 2026 10:23:19	One Outlook Web	9199bf20-a13f-4107-85dc	["interactiveUser"]	OAuth2:Token	0
☐	> Jun 1, 2026 10:22:24	One Outlook Web	9199bf20-a13f-4107-85dc	["interactiveUser"]	OAuth2:Token	0
☐	> Jun 1, 2026 10:22:25	One Outlook Web	9199bf20-a13f-4107-85dc	["interactiveUser"]	OAuth2:Token	0
☐	> Jun 1, 2026 10:23:25	Office 365 Exchange Onlin	00000002-0000-0ff1-ce00-	["nonInteractiveUser"]	OAuth2:Token	0

7. Conclusion & Recommendations

Capturing and retaining Entra ID sign-in logs is critical for detecting anonymized access attempts via the TOR network. The implemented KQL detection rule provides real-time, automated coverage against this threat vector using authoritative, dynamically updated data from the TOR Project itself. This detection is a crucial part of the investigation done via our model, and the determination of a potential internal phishing.